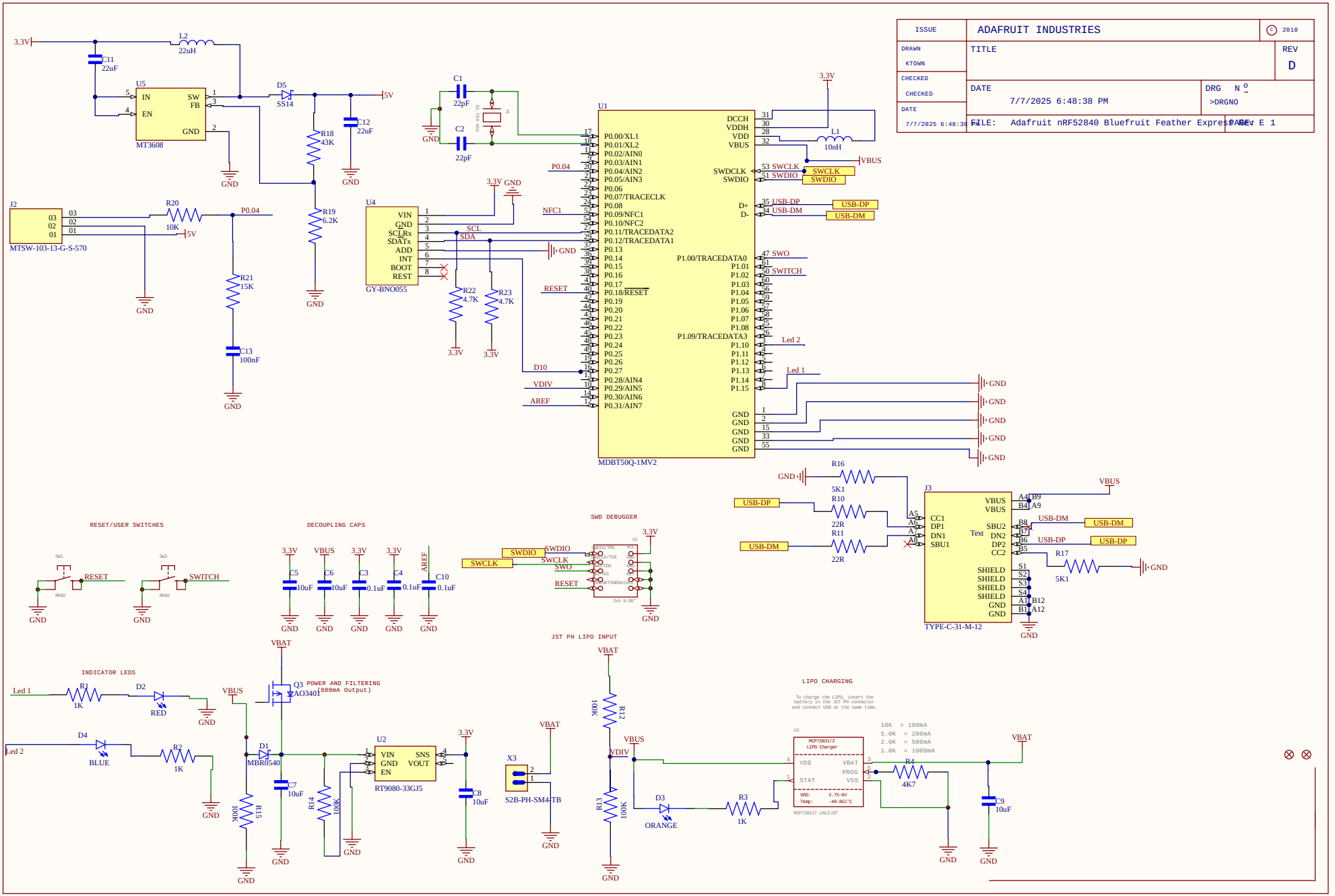
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FILE: Adafruit nRF52840 Bluefruit Feather Express.SCH
SHEET 1
RESET/USER SWITCHES
DECOUPLING CAPS
SWD DEBUGGER
INDICATOR LEDS
JST PH LIPO INPUT
LIPO CHARGING
POWER AND FILTERING
(600mA Output)
MT3608
MTSW-103-13-G-S-570
GY-BNO055
MDBT50Q-1MV2
TYPE-C-31-M-12
RT9080-33GJ5
MCP73831/2
LIPO Charger
To charge the LIPO, insert the battery in the JST PH connector and connect USB at the same time.
10K = 100mA
5.0K = 200mA
2.0K = 500mA
1.0K = 1000mA

Comment	Description	Designator	Footprint	LibRef	Quantity
22pF	22 pF Æ9% 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)	C1, C2	CAPC1608X90	C0603JNPO9BN220	2
0.1uF	0.1 ÂµF Æ±10% 16V Ceramic Capacitor X7R 0603 (1608 Metric)	C3, C4, C10	CAPC1608X90N	C0603C104K4RAC	3
10uF	10 ÂµF Æ±10% 16V Ceramic Capacitor X5R 0805 (2012 Metric)	C5, C6, C7, C8, C9	CAPC2012X88N	C0805C106K4PACTU	5
22uF	22 µF ±20% 25V Ceramic Capacitor X5R 1206 (3216 Metric)	C11, C12	CAPC3216X259N	12063D226MAT2A	2
100nF	Capacitor MLCC 0603 100nF 50V	AVX 0603C104K4T2A	C13	CAPC1608X90N	0603SC104K4T2A	1
MBR0540	Diode Schottky 40 V 500mA Surface Mount SOD-123	D1	DIO_MBR0540-TP	MBR0540-TP	1
RED	LED Uni-Color Green 4 Pin Chip 0805 T/R	D2	LEDC2012X70N	597-3305-607F	1
ORANGE	LED Uni-Color Green 4 Pin Chip 0805 T/R	D3	LEDC2012X70N	597-3305-607F	1
BLUE	LED Uni-Color Green 4 Pin Chip 0805 T/R	D4	LEDC2012X70N	597-3305-607F	1
SS14	Diode Schottky 40 V 1A Surface Mount DO-214AC (SMA)	D5	SMA	SS14	1
FIDUCIAL_1MM	Fiducial Alignment Points Various fiducial points for machine vision alignment.	FID1, FID2	FIDUCIAL_1MM	FIDUCIAL_1MM	2
MTSW-103-13-G-S-570	Connector Header Through Hole 3 position 0.100 (2.54mm)	J2	SAMTEC_MTSW-103-13-G-S-570	MTSW-103-13-G-S-570	1
TYPEC-31-M-12	USB Connectors 24 Receptacle 1 8.94* 7.3mm RoHS	J3	HRO_TYPEC-31-M-12	TYPEC-31-M-12	1
10nH	FIXED IND 10NH 600MA 100 MOHM	L1	INDC2015X152N	LQW2BAS2ONG00L	1
22uH	Inductor with Inductance: 22uH Tol: +/-20%, Package: 0805 (2012)	L2	INDC2012X145N	LQM21DH220M70L	1
AO3401	P-Channel 30 V 4A (Ta) 1.4W (Ta) Surface Mount SOT-23-3	Q3	SOT23-3	AO3401	1
1K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R1, R2, R3	RES_ERJU03J5R6V	ERJU03J5R6V	3
4K7	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R4	RES_ERJU03J5R6V	ERJU03J5R6V	1
22R	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R10, R11	RES_ERJU03J5R6V	ERJU03J5R6V	2
100K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R12, R13, R14, R15	RES_ERJU03J5R6V	ERJU03J5R6V	4
5K1	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R16, R17	RES_ERJU03J5R6V	ERJU03J5R6V	2
43K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R18	RES_ERJU03J5R6V	ERJU03J5R6V	1
6.2K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R19	RES_ERJU03J5R6V	ERJU03J5R6V	1
10K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R20	RES_ERJU03J5R6V	ERJU03J5R6V	1
15K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R21	RES_ERJU03J5R6V	ERJU03J5R6V	1
4.7K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R22, R23	RES_ERJU03J5R6V	ERJU03J5R6V	2
RKB2	SMT Tact Switches RKB2 https://www.digikey.com/short/md27yrz5	SW1, SW2	BTN_RKB2_4.6X2.8	SWITCH_TACT_SMT_RKB2	2
MDBT50Q-1MV2	802.15.4, Bluetooth Bluetooth v5.4, ZigbeeÂ® transceiver Module 2.36GHz – 2.5GHz Integrated, Chip Surface Mount	U1	XCVR_MDBT50Q-1MV2	MDBT50Q-1MV2	1
RT9080-33GJ5	Linear Voltage Regulator IC Positive, Fixed 1 Output 600mA TSOT-23-5	U2	SOT95P282X100-5N	RT9080-33GJ5	1
MCP73831T-2ACI/OT	MCP73831/2 LiPO Charger VDD: 3.75 - 6V Temp -40 - 85Â°C Programmable Charge Rate: 15-500mA NOTE: STAT is a tri-state logic output on the MCP73832. SOT23-5 with 4.2V output and tri-state logic on STAT – MCP73831T-2ACI/OT	U3	SOT23-5	MCP73831/2	1
GYBINO055	GYBINO055	U4		GYBINO055	1
MT3608	High Efficiency 1.2MHz 2A Step Up Converter	U5	SOT95P280X145-6N	MT3608	1
2x5 0.05"	Serial Wire (SWD) Connector Standard 0.05" 10-pin connector is for use with the Cortex-M and Cortex-A SWD (serial wire debugger) interface (SW-DP) Connectors 2x05_1.27MM_SMT - Inexpensive surface-mount header with bare pins, compatible with standard 10-pin SWD cables (4UCON: 16846) 2x05_1.27MM_BOX_NOPOSTS - Polarised surface-mount box (4UCON with caps: 19735, without caps: 15117 2x05_1.27MM_BOX_POSTS - Polarised surface-mount box with mounting holes (4UCON: 20310)	X2	2X05_1.27MM_BOX_POSTS	JTAG-CORTEXBOXPOSTS	1
S2B-PH-SM4-TB	Connector Header Surface Mount, Right Angle 2 position 0.079 (2.00mm)	X3	JST_S2B-PH-SM4-TB	S2B-PH-SM4-TB	1
32.768 KHz	Low cost SMT crystals, no capacitors included http://www.ladyada.net/library/eagle	Y2	XTAL3215	XTAL-3.2X1.5	1